Nonlinear Practice 2

Module 8 - Nonlinear Pharmacokinetics · Daily practice sheet · No date

Name _____ Date _____

1a 1. A 75-kg, 32-year-old female has been taking 200 mg phenytoin daily. Because her average phenytoin plasma concentration was only 6 mg/L, her dose was increased to 350 mg/day. The steady-state average phenytoin concentration was then found to be 21 mg/L. (Volume of distribution = 0.75 L/kg and F = 1) a. Calculate the patient's Vmax and KM. What is the KM?

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Question 2 What is the difference between linear and non-linear pharmacokinetics?

A. Nonlinear processes affect absorption only, never elimination
B. Linear parameters stay fixed with dose; nonlinear ones change with dose
C. In nonlinear pharmacokinetics the AUC rises in direct proportion to the dose
D. Linear pharmacokinetics is zero order and nonlinear pharmacokinetics is first order

Question 3 What changes in t½ and Cl_T are usually observed with increased concentrations of drugs that exhibit nonlinear pharmacokinetics?

A. t½ increases and Cl_T increases

B. Neither changes, because Vmax and KM are constants

C. t½ decreases and Cl_T increases

D. t½ increases and Cl_T decreases